Evacuate- & Fill Process

Repair Cycle and Service Bleed; EBC460 ABS for SGMW GP30

Revision: 0.1

Sep 22. 2014

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ADVANCED THINKING / SMART THINKING / GREEN THINKING

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1. Revision Log

Version	Date	Author	Section affected	Changes
0.1	Sep. 22. 2014	Rudy Wang	all	Initial

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3. Definition

- Evacuate and Fill Cycle

  is intended to be used to fill dry brake systems at customer assembly line

- Repair Cycle

  is intended to be used to fill “ wet” brake systems in case of Evacuate and Fill Cycle had failed at customer assembly line

- Service Bleed Cycle

  is intended to be used in a Workshop to change brake fluid, brake components or “prefilled” Service – ABS - units

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4. Introduction for Evac- & Fill and Repair cycle

4.1 Diagnosis via CAN

4.2 Conditions for I/O control

- Only in EOL vehicle manufacturer session (for v < 3kph; entry condition)
- No active control algorithm (ABS / Degas pulse…)
- Remember to activate Tester Present to keep module in EOL session
- I/O command expires after 4seconds, can be retriggered
- I/O control will be inactive if vehicle speed exceeds v = 10kph
- No under- or over-voltage (12V – 13,5V; 10,5A)
- No valve errors or wheel speed-sensor fault present

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5. Diagnostic Communication

Process byte

Byte	Read Evacuate and Fill Request: Description / Resolution	Value
1	request byte	$22
2	Data LID MSB Byte	**$63**
3	Data LID LSB Byte	**$00**

Byte	Write Evacuate and Fill Response: Description / Resolution	Value
1	reply byte	$6E
2	Data LID MSB Byte	**$63**
3	Data LID LSB Byte	**$00**

Byte	Read Evacuate and Fill Response: Description / Resolution	Value
1	reply byte	$62
2	Data LID MSB Byte	**$63**
3	Data LID LSB Byte	**$00**
4	Evacuate and Fill Status Filling Not Successful Filling Successful Delivery State	$xx = $00 $7F $FF

Byte	Write Evacuate and Fill Request: Description / Resolution	Value
1	request byte	$2E
2	Data LID MSB Byte	**$63**
3	Data LID LSB Byte	**$00**
4	Evacuate and Fill Status Filling Not Successful Filling Successful Delivery State	$xx = $00 $7F $FF

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5.1 Diagnostic Communication

I / O control / Valve activation

Byte	IO-Control Request (ABS/ESP): Description / Resolution	Value
1	request byte	$2F
2	Data Id MSB Byte	**$D3**
3	Data Id LSB Byte	**$01**
4	output control parameter Return Control to ECU Short Term Adjustment	$xx = $00 $03
5	BIT 7 reserved BIT 6 reserved BIT 5 Pump motor BIT 4 FS relais NO access via I/O Ctrl! BIT 3 TC/VSC w.lamp BIT 2 TC /VSC info Lamp BIT 1 DRP W Lamp BIT 0 ABS W Lamp	truth table: bit = 0 OFF bit = 1 ON
6	Solenoid valves Control byte BIT 7 rr iso BIT 6 rr dump BIT 5 lr iso BIT 4 lr dump BIT 3 rf iso BIT 2 rf dump BIT 1 lf iso BIT 0 lf dump	truth table: bit = 0 valve not energized bit = 1 valve energized
7	TC valves Control byte BIT 7… 4 Reserved (Always 0) BIT 3 TC iso 2 RFLR BIT 2 TC Iso 1 LFRR BIT 1 TC supply 2RFLR BIT 0 TC supply 1 LFRR	truth table: bit = 0 not energized bit = 1 energized
8	Reserved (Always 0)	
9-12	IO Control Mask Bytes (bit map same as the bytes 5-8)	truth table: bit = 1 valve should be controlled bit = 0 valve should not be controlled

Byte	IO-Control Response (ABS/ESP): Description / Resolution	Value
1	Reply byte	**$6F**
2	Data Id MSB Byte	**$D3**
3	Data Id LSB Byte	**$01**
4	Pump Motor and Lamps Info Bitmap same as that in request	
5	Solenoid valves Control byte Bitmap same as that in request	
6	TC value Control Status	
7	Reserved (Always 0)	

Command definitions and details are contained in the document: Diagnostic_Proposal_.doc

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6. Recommended Evacuate and Fill Cycle TRW

- Connect Evacuate and Fill machine to reservoir of master cylinder
- Connect power supply to ECU
- Connect diagnostic line to ECU- or OBD- connector
- Start communication via diagnostic line
- Start Evacuate and Fill Cycle
- Disconnect diagnostic line and check pedal travel after some time
- If the pedal travel is too long, perform the repair cycle including the pedal travel measurement

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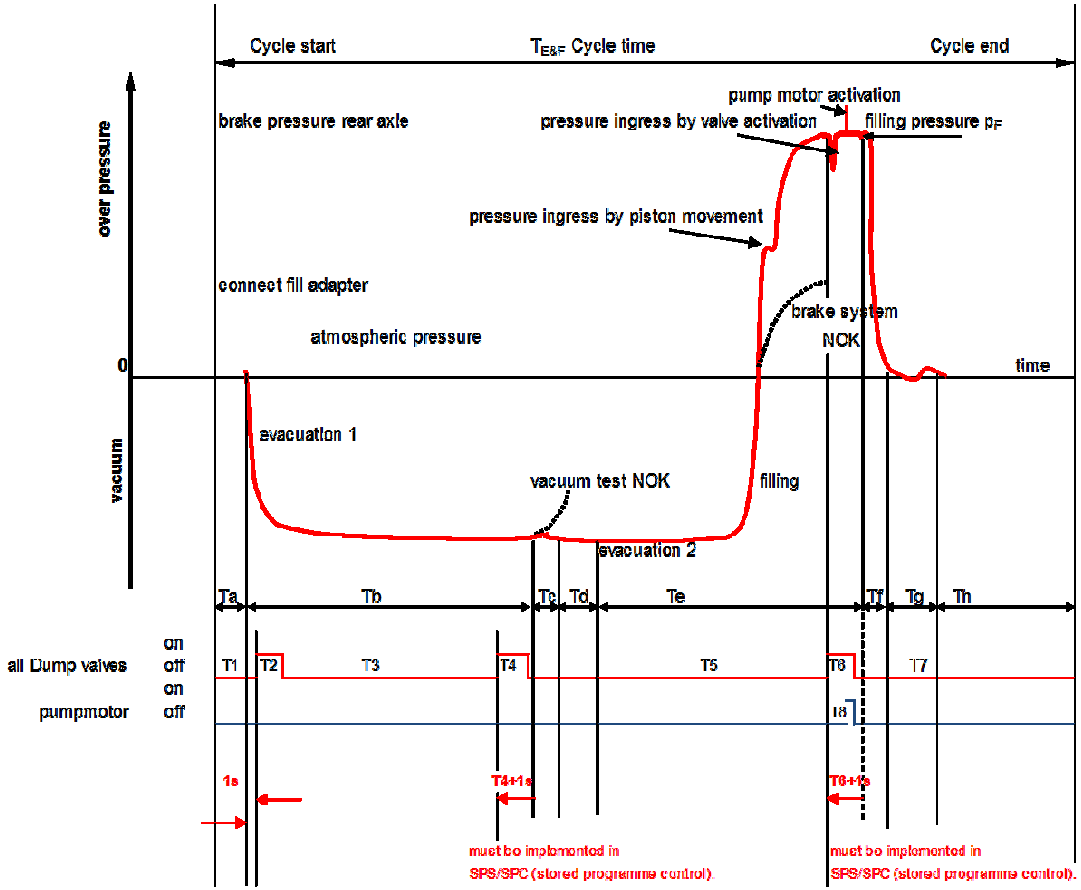
7. Evacuate and Fill Cycle TRW EBC460 ABS only



Time table

Ta - prepare & clamp adaptor		s
Tb - evacuation 1	≥	s
Tc - leak test		s
Td - evacuation 2		s
		s
Te - filling time	≥	s
Tf - relieve pressure		s
Tg - levelling		s
Th - rest(handling, console reverse)		s
$T_{E\&F}$ evac-& fill time		s
p_F - filling pressure	≥	4 bar

Valve activation

Dump LF, RF, LR, RR

T1 - electrical off	s
T2 - electrical on	4 s
T3 - electrical off	s
T4 - electrical on	4 s
T5 - electrical off	s
T6 - electrical on	4 s
T7 - electrical off	s

Pump motor activation

optional

T8 - electrical on	1 s

Valve activation:

$2F $D3 $01 $03 $00 $55 $00 $00 $00 $55 $00 $00

additional pump motor start:

$2F $D3 $01 $03 $20 $55 $00 $00 $20 $55 $00 $00

T2 – Valve activation 1 s after start evacuation.

Tb_{End} / Td_{End} < 5 mbar at rear axle brake $T4_{Start}$ = Tb-T4-1s; $T4_{Start}$ must be implemented in SPS/SPC (stored programme control)!

$T6_{Start}$ = Valve activation if brake system completely filled. $T6_{Start}$ = Te-T6-1s must be implemented in SPS/SPC (stored programme control)!

Ta up to Th (incl.TE&F) are resulting of the machine parameter (evac & fill machine) and the filling volume of the vehicle.

Valve activation corresponds to: Dump valve ABS left front (LF) / right front (RF) / left rear (LR) / right rear (RR)..

T2, T4, T6 : current I ~ 10,5 A (U = 12V - 13,5V)

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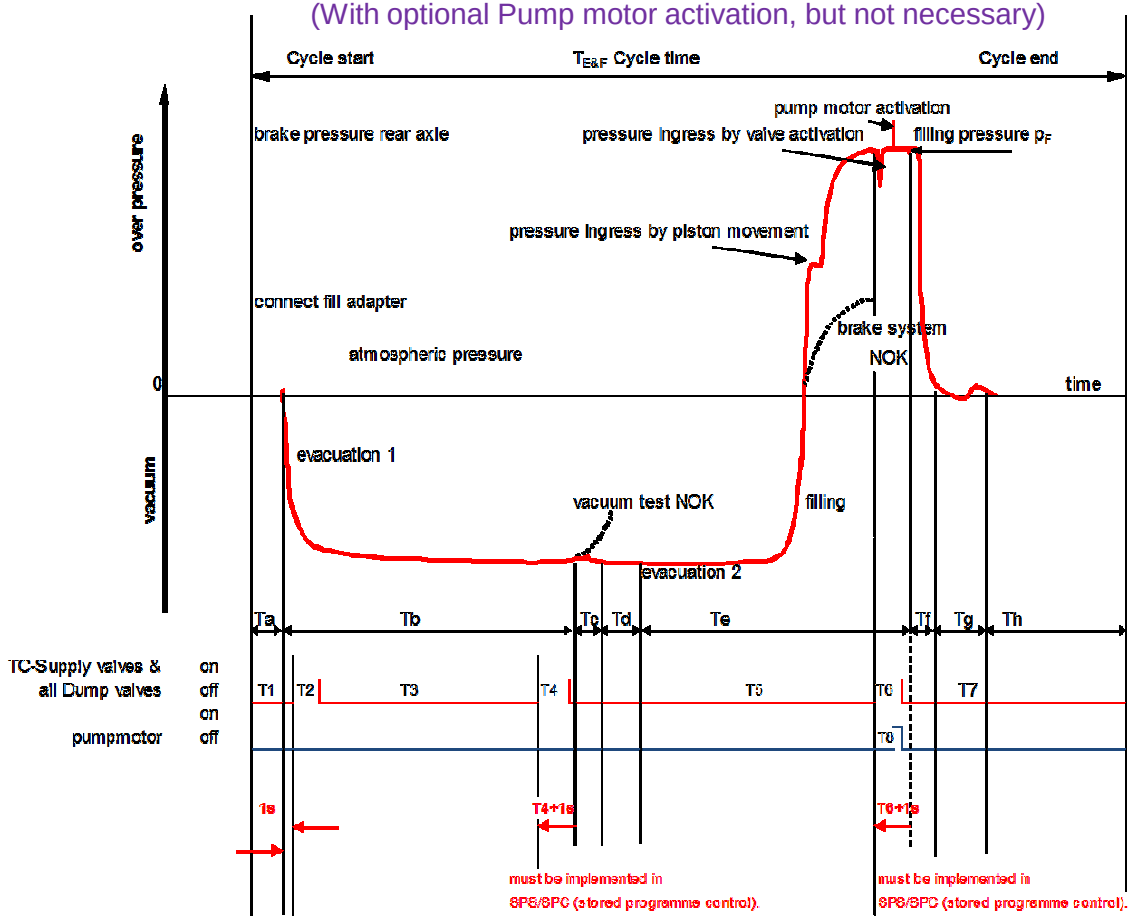
8. Evacuate and Fill Cycle TRW EBC460 ESC

(With optional Pump motor activation, but not necessary)



Time table

Ta - prepare & clamp adaptor		s
Tb - evacuation 1	≥	s
Tc - leak test		s
Td - evacuation 2		s
		s
Te - filling time	≥	s
Tf - relieve pressure		s
Tg - levelling		s
Th - rest(handling, console reverse)		s
$T_{E\&F}$ evac-& fill time		s
p_F - filling pressure	≥	4 bar

Valve activation

Dump LF, RF, LR, RR & TC- Supply1/2

T1 - electrical off	s
T2 - electrical on	4 s
T3 - electrical off	s
T4 - electrical on	4 s
T5 - electrical off	s
T6 - electrical on	4 s
T7 - electrical off	s

Pumpmotor activation

optional

T8 - electrical on	1 s

Valve activation:

$2F $D3 $01 $03 $00 $55 $03 $00 $00 $55 $03 $00

additional pump motor start:

$2F $D3 $01 $03 $20 $55 $03 $00 $20 $55 $03 $00

T2 ? Valve activation 1 s after start evacuation.

Tb_{End} / Td_{End} < 5 mbar at rear axle brake $T4_{Start}$ = Tb-T4-1s; $T4_{Start}$ must be implemented in SPS/SPC (stored programme control)!

$T6_{Start}$ = Valve activation if brake system completely filled. $T6_{Start}$ = Te-T6-1s must be implemented in SPS/SPC (stored programme control)!

Ta up to Th (incl.TE&F) are resulting of the machine parameter (evac & fill machine) and the filling volume of the vehicle.

Valve activation corresponds to: Supply valve 1 / Supply valve 2 (ESP) and Dump valve ABS left front (LF) / right front (RF).
and Dump valve ABS left rear (LR) / right rear (RR).

T2, T4, T6 : current I ~ 10,5 A (U = 12V - 13,5V)

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9. Test Procedure for Evac- & Fill

- Mounting the EBC 460 unit with a complete brake system in a vehicle, or simulating a original vehicle installation at a test bench including booster/pedal, hoses, piping and brake system
- Install additional measurement equipment (vacuum and pressure transducer and pedal travel sensor)
- Pre Vacuum Leak Test – Vacuum at rear caliper should be less than 5mbar after specified evacuation time and no leakage, to get acceptable Evacuate and Fill results
- Evacuate and Fill with de-gassed brake fluid according recommended Evacuate and Fill cycle
- Bleed check per pedal travel measurement

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10. Recommended Repair Cycle (Plant Rework Station)

- Connect Evacuate and Fill rework machine to reservoir of master cylinder
- Connect diagnostic line and power supply to OBD - connector
- Start communication via diagnostic line
- Start Repair Cycle
- After Repair Cycle has finished, check pedal travel
- If the pedal travel is too long, use the repair cycle once again

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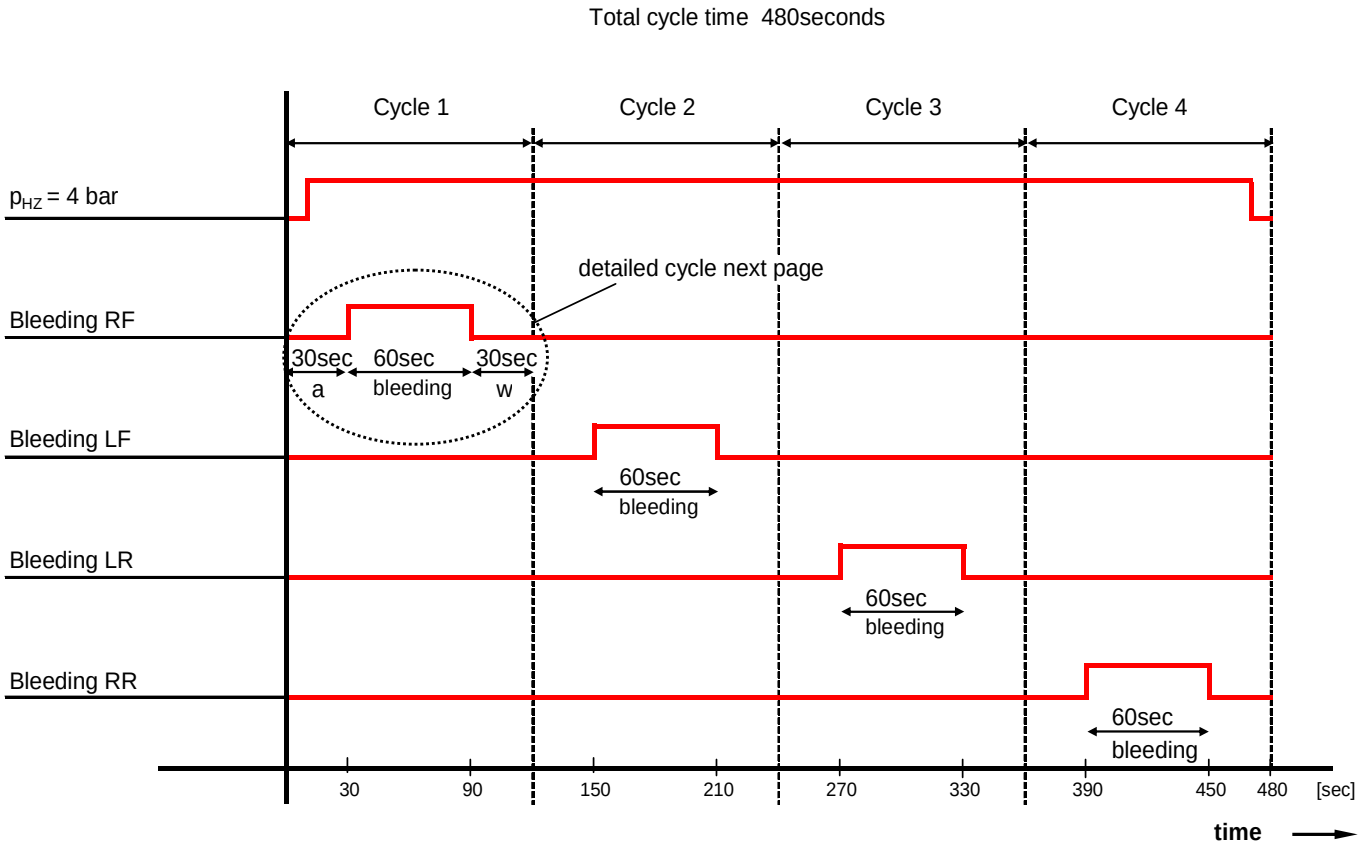
10.1 Repair Cycle

(Is intended to be used in case of Evacuate and Fill Cycle failed at customer assembly line)



Procedure:

1. Connect bleeder ball with 4 bar pressure.
2. Start repair cycle.
3. Start to open bleeder screw and keep it open until all air has been removed.

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10.1 Repair Cycle

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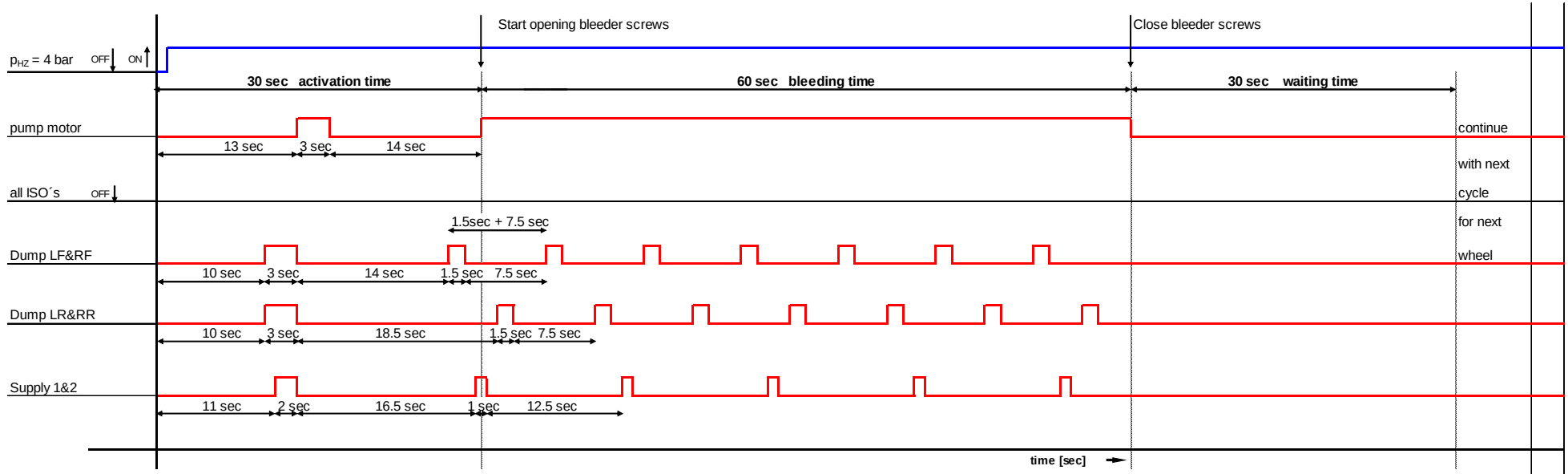

Procedure:

Supply voltage U_{max} 10.5 to 12.0 Volts.
Min. waiting time before start next cycle is 30 seconds.

1. Connect bleeder ball with 4 bar pressure.
2. Start repair cycle.
3. Start to open bleeder screw and keep it open until all air has been removed.

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11. Recommended Service Bleed *(with prefilled service units only)*

- Connect Brake -Fill machine to reservoir of master cylinder and apply pressure to brake system (min 2 bar)
- Start to open bleeder screw LF and keep it open until all air has been removed.
- Start to open bleeder screw RF and keep it open until all air has been removed.
- Start to open bleeder screw RR and keep it open until all air has been removed.
- Start to open bleeder screw LR and keep it open until all air has been removed.
- After Service Bleed has finished, check pedal travel
- If the pedal travel is too long, start Service bleed once again

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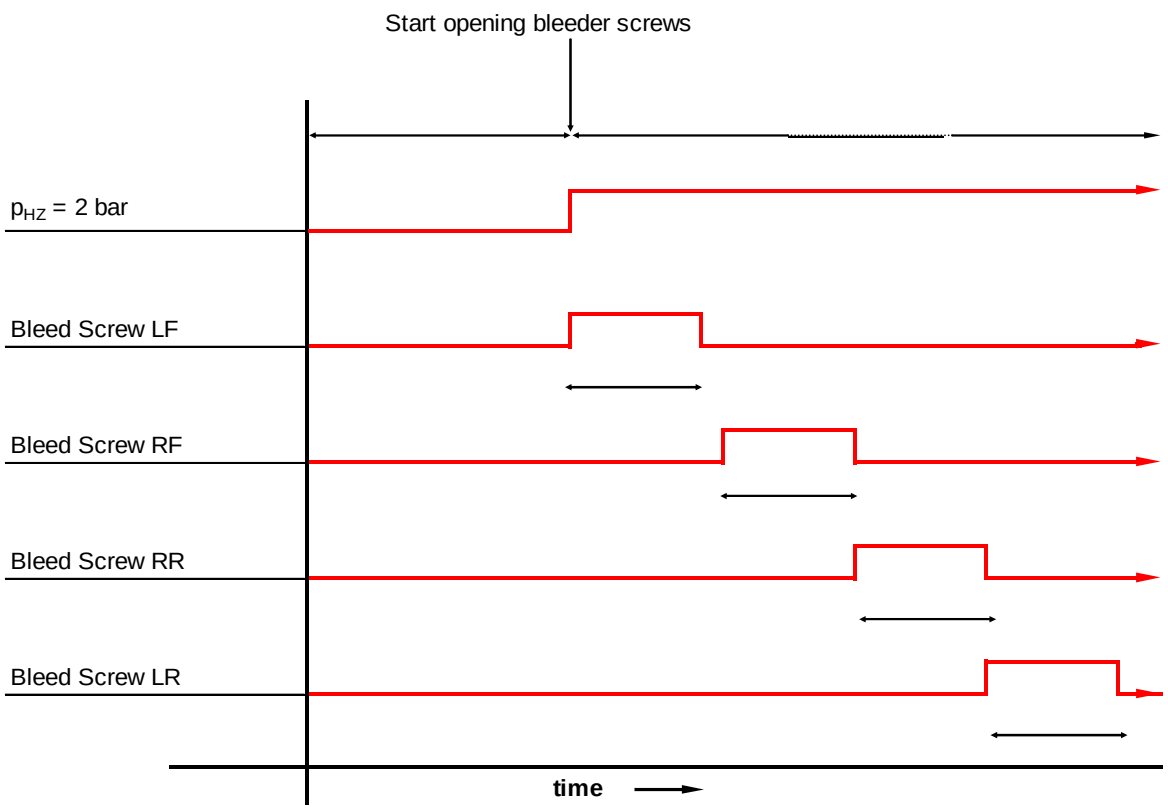
11.1 Service Bleed Cycle *(is intended to be used in a Workshop to change brake fluid, brake components or prefilled ESC units)*



Procedure:

1. Connect bleeder ball with reservoir and apply pressure to brake system.
2. Start to open bleeder screw and keep it open until all air has been removed.

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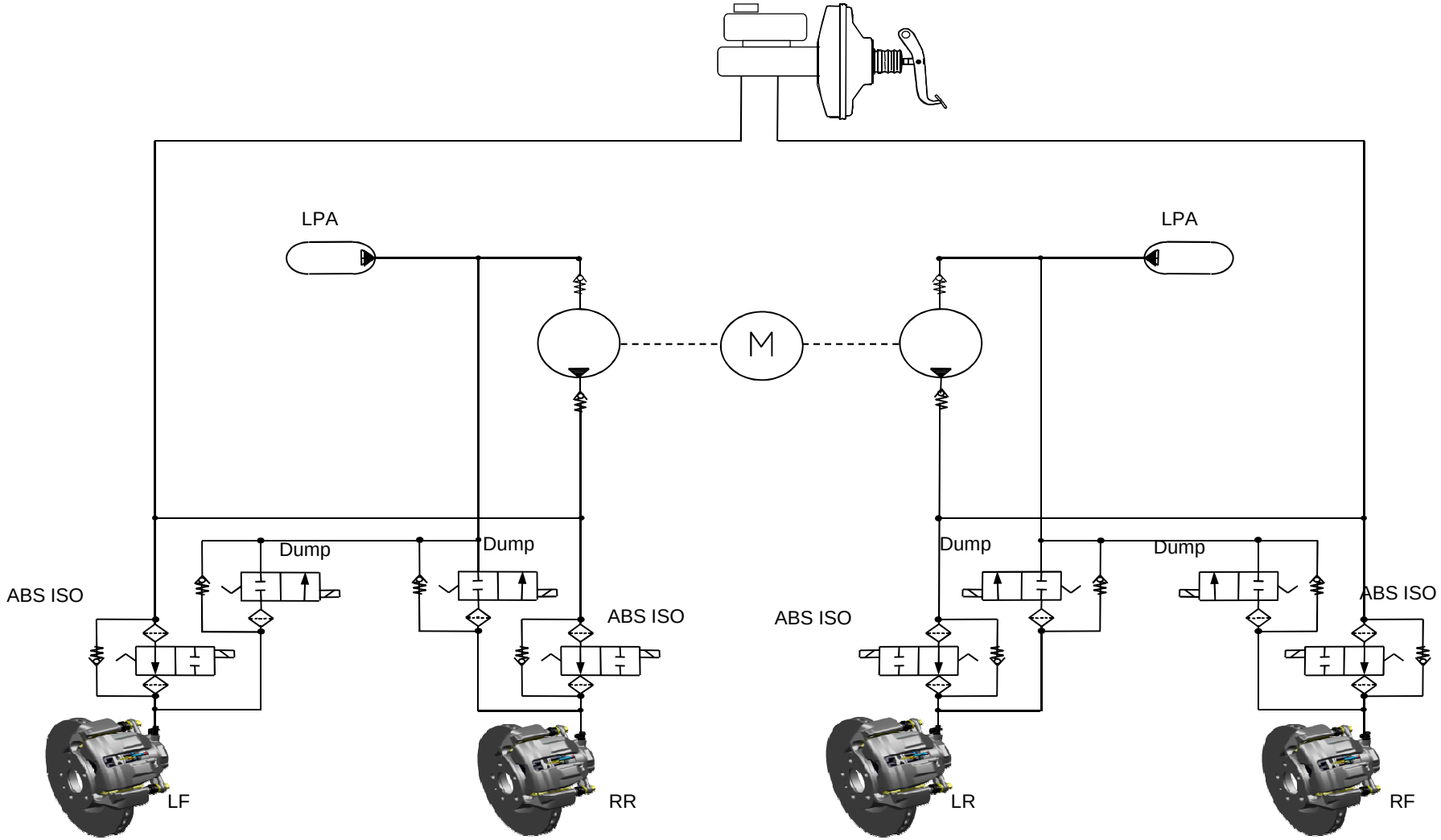
12. Hydraulic diagram ABS



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